

Course Contents and Tentative Class Schedule Table-I (subject to change)

Week No		Topics	Book Name	Chapter No	Exam/Quiz/Assignment	
					Category	Time Line for Result Submission
1		PARTIAL DERIVATIVES: Function of two or more variables	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	13	NA	NA
		Limit and continuity of function of two variables	Same as Above	13	NA	NA
2		Partial derivatives	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	13	Home Work/ Assignment 1 (Week 2/Week3)	Within 7 Days
		Differentiability and chain rule of function of two variable	Same as Above	13	NA	NA
3		Directional derivatives Gradient for function of two variables	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	13	NA	NA
		Differentiability Directional derivatives Gradient for function of three variables	Same as Above	13	Quiz 1 (Week 3/Week4)	Within 7 Days
4		Tangent planes and normal vectors Total differential for function of two variables	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	13	NA	NA
		Maxima and minima of function of two variables	Same as Above	13	NA	NA
5		MULTIPLE INTEGRALS: Double integral over rectangular region	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc..	14	Home Work/ Assignment 2 (Week 5/Week6)	Within 7 Days
		Double integral over non-rectangular region.	Same as Above	14	NA	NA
6		Reversing the order in Double integrals, Area by Double integral	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	14	NA	NA
		Double integral in polar coordinates.	Same as Above	14	Quiz 2 (Week 6/Week7)	Within 7 Days
7		Conversion of Double integrals from rectangular to polar coordinates.	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	14		
		Introduction to Triple integrals. Triple integral over rectangular region.	Same as Above	14		

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8		Triple integral over non-rectangular region.	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	14		
		Triple integral in cylindrical coordinates Triple integral in spherical coordinates	Same as Above	14		
9		Mid Term Exam Week			Midterm	Within 5 Days
10		VECTOR CALCULUS: Vector fields, Inverse -square field, Gradient fields, Conservative fields and Potential functions, Divergence and Curl.	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	15	Home Work/ Assignment 3 (Week10/Week11)	Within 7 Days
		Divergence and Curl. The del operator, The Laplacian	Same as Above	15	NA	NA
11		Line integrals	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	15	Quiz 3 (Week 11/ Week12)	Within 7 Days
		Work as a Line integral, Fundamental theorem for line integrals	Same as Above	15		
12		Conservative field test, finding Potential functions	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	15	Home Work/ Assignment 4 (Week12/Week13)	Within 7 Days
		Green’s theorem, Work and Area using Green’s Theorem, Parametric surfaces and their areas	Same as Above	15		
13		Parametric surfaces and their areas	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	15	Quiz 4 (Week 13/Week14)	Within 7 Days
		Surface integrals.	Same as Above	15	NA	NA
14		Application of surface integrals: Flux	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	15		
		Divergence theorem, finding flux using the divergence Theorem	Same as Above	15		
15		Stokes’ theorem, calculating work using Stokes’ Theorem	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	15		

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		FOURIER SERIES: Euler formulas Fourier series of the functions having period $p = 2\pi$	Same as Above	15		
16		Functions of any Period $P=2L$	“Calculus Early Transcendentals”, by Howard Anton, 10th edition, John Wiley and sons, Inc.	15		
		Transition from period 2π to any period $2L$	Same as Above	15		
17		Final Exam Week			Final Term	Within 3 Days
18		Final Exam Week				